

Kishoreganj University (KiU)
3rd Year 1st Semester B. Sc. (Engg.) First In-course Examination-2025
Department of Computer Science and Engineering
CSE 3107: Information System Design (3 Credits)

Figures shown in the right margin indicate full marks.
Answer all questions

Total Marks: 25
Time: 1 hour

1. a) Suppose you are developing a prototype for a ticket booking system. Users complain it's too slow, but speed was never mentioned as a requirement during the elicitation phase. Which type of requirement was overlooked here and how could this have been avoided? 3

b) You are developing a banking app and a week before launch, you noticed that it still has known bugs in the transaction module. Management started saying to release it now and fix it in the next update. Why could this be a problem, and what would you do in such a situation? 3
2. a) In a drawing app, once a user starts a "Draw Polygon" action, they cannot switch tools or stop midway. The only option they have is to complete the shape. If they make a mistake, they must start over from scratch. What is problematic in this approach? How would you redesign it? 3

b) Suppose your team finalized the requirements for a new banking app but skipped creating usage scenarios or diagrams. During development, you discovered major gaps in transaction workflows. What validation step was missed in this case and why is it important? 3

c) Suppose you built an in-house inventory tracking system and only gathered requirements from junior developers because they were available. After delivery, the CEO rejected it, saying it won't solve the company's real inventory problems. What problem occurred here, and in which phase of development should it have been addressed? 3
3. LexAssist is developing an intelligent legal support system to help users access legal information and get answers to their queries quickly. Users can submit questions about various legal topics such as contracts, property law, or employment rights. The system provides relevant legal documents, explains key terms in simple language, and suggests possible next steps. Legal advisors can review user queries, provide detailed responses, and update the knowledge base with new precedents or regulations. Administrators manage user roles, monitor query trends, and ensure the system's content stays up-to-date with current laws. The system also allows users to schedule consultations with lawyers and track the status of their cases. 10

a) From this scenario, identify at least five use cases and create a use case diagram illustrating the actors and their interactions.

b) Develop a class diagram outlining the main classes, their attributes, methods, and relationships, including associations and multiplicities.